

Section C

10 **(short)** Assume that charge of Q is uniformly distributed on a sphere of radius R .

(a) Derive expressions for the electric field inside and outside the sphere. [4]

(b) Derive an expression for the capacitance of the sphere (the potential at infinity is zero). [4]

(c) With $\epsilon_0 = 8.854 \times 10^{-12} \text{ F m}^{-1}$, estimate the radius of the sphere if the capacitance is 1 pF. [2]

Write down each step and state clearly any approximations made.